

Week 3 In-Class Worksheet

Probability Distributions and Sampling Distributions

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Binomial, Normal, Sampling Distributions, and Confidence Intervals

Instructions

Work with a partner to solve the following problems. Show all your work. You may use R to verify your answers, but demonstrate your understanding of the concepts by showing calculations.

1 Problem 1: Binomial Distribution

A new antibiotic has an 80% success rate in treating a particular bacterial infection. A clinic treats 15 patients with this antibiotic.

Part A: What is the probability that exactly 12 patients are successfully treated?

Given: $n = 15$, $p = 0.80$, find $\Pr(X = 12)$

Formula: $\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$

Part B: What is the probability that at least 13 patients are successfully treated?

Hint: $\Pr(X \geq 13) = \Pr(X = 13) + \Pr(X = 14) + \Pr(X = 15)$

Part C: Calculate the expected number of successful treatments and the standard deviation.

Expected value: $\mu = np$

Standard deviation: $\sigma = \sqrt{np(1-p)}$

Part D: Use R to verify your answers. Write the R code you would use:

2 Problem 2: Normal Distribution

Adult systolic blood pressure is normally distributed with mean 125 mmHg and standard deviation 18 mmHg.

Part A: What proportion of adults have blood pressure above 140 mmHg (hypertension)?

Step 1: Standardize using $Z = \frac{X - \mu}{\sigma}$

Step 2: Find $\Pr(Z > z)$ using the standard normal table or R

Part B: What proportion have blood pressure between 110 and 140 mmHg?

Hint: $\Pr(110 < X < 140) = \Pr(Z < z_2) - \Pr(Z < z_1)$

Part C: Find the blood pressure value that represents the 90th percentile.

Step 1: Find z such that $\Pr(Z < z) = 0.90$

Step 2: Convert back to original scale: $X = \mu + z\sigma$

Part D: Write the R code to solve parts A, B, and C:

3 Problem 3: Sampling Distribution and Central Limit Theorem

A population of cholesterol levels has mean $\mu = 200$ mg/dL and standard deviation $\sigma = 40$ mg/dL. A sample of $n = 64$ individuals is randomly selected.

Part A: What is the distribution of the sample mean $\bar{X}$?

By the Central Limit Theorem: $\bar{X} \sim N(\mu_{\bar{X}}, \sigma_{\bar{X}}^2)$

Mean of $\bar{X}$: $\mu_{\bar{X}} =$

Standard error of $\bar{X}$: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} =$

Part B: What is the probability that the sample mean is greater than 210 mg/dL?

Step 1: Standardize: $Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$

Step 2: Find probability

Part C: What is the probability that the sample mean is between 195 and 205 mg/dL?

Part D: How would your answers change if $n = 16$ instead of $n = 64$? Would that probability increase or decrease? Explain why.

4 Problem 4: Confidence Intervals

A clinical trial measures the reduction in pain scores for 20 patients receiving a new medication. The sample mean reduction is $\bar{x} = 4.5$ points, with sample standard deviation $s = 1.8$ points.

Part A: Calculate a 95% confidence interval for the true mean pain reduction.

Step 1: Determine what critical value to use. Should you use z or t ? Why?

Step 2: Find the critical value.

Degrees of freedom = $n - 1 =$

Critical value: $t_{0.975, df} =$ (use R: `qt(0.975, df)`)

Step 3: Calculate the standard error: $SE = \frac{s}{\sqrt{n}}$

Step 4: Calculate the margin of error: $ME = t_{critical} \times SE$

Step 5: Construct the confidence interval: $\bar{x} \pm ME$

Part B: Interpret the confidence interval in context. Write a complete sentence explaining what this interval means.

Part C: If you wanted a 99% confidence interval instead, would it be wider or narrower than the 95% interval? Explain.

5 Problem 5: Integration Problem

A pharmaceutical company is testing a new cholesterol-lowering drug. In the clinical trial:

- 100 patients receive the drug
- Cholesterol reduction follows a normal distribution
- Sample mean reduction: $\bar{x} = 28$ mg/dL
- Sample standard deviation: $s = 12$ mg/dL

Part A: Calculate a 95% confidence interval for the true mean cholesterol reduction.

Part B: Based on your confidence interval, is there strong evidence that the drug reduces cholesterol by more than 25 mg/dL on average? Explain your reasoning.

Part C: If individual cholesterol reductions are normally distributed with the sample statistics above, approximately what proportion of patients will experience a reduction of at least 40 mg/dL?

Hint: Use the sample statistics as estimates of population parameters.

Part D: Explain the difference between the questions in Parts B and C. Why do we use different approaches (confidence interval vs. normal probability)?

Reflection Questions

1. What is the key difference between a standard deviation and a standard error?
2. When should you use a t -distribution instead of a normal distribution for confidence intervals?
3. How does sample size affect the width of a confidence interval?